Full Stack Assignment: Timesheet Management System

Objective: Build a production-quality Timesheet Management system using .NET 8 and Angular. The system should allow employees to submit timesheets against assigned project codes and managers to manage projects, approve hours, and generate reports.

Roles

- Employee: Submit weekly timesheets against assigned project codes.
- Manager: Create project codes, assign them to employees, approve or reject timesheets, and view reports.

Project Code Management (Manager)

- Create, update, activate, and deactivate project codes.
- Each project code must include: Project Code, Project Name, Client Name, Billable Flag, Status.

Project Assignment (Manager)

- Assign one or more project codes to employees.
- Define assignment start and end dates.
- Employees can submit timesheets only for assigned and active project codes.

Timesheet Submission (Employee)

- Submit weekly timesheets with date, project code, hours worked, and description.
- Maximum 24 hours per day.
- No duplicate entries for same project code and date.
- Timesheet lifecycle: Draft → Submitted → Approved / Rejected.

Timesheet Approval (Manager)

- View pending timesheets submitted by employees.
- Approve or reject timesheets.
- Rejection requires mandatory comments.

Reports (Manager)

- Employee-wise hours summary.
- Project-wise hours summary.
- Billable vs non-billable hours.
- Date range based filtering.
- All aggregations must be done using EF Core and LINQ queries.

Backend Requirements (.NET 8)

- RESTful Web API using .NET 8.
- Entity Framework Core with code-first approach.
- AutoMapper for entity-DTO mapping.
- Dependency Injection for repositories and services.
- Design Patterns: Factory, Strategy, Decorator.
- Unit testing using NUnit and Moq.

Frontend Requirements (Angular)

- Feature-based module architecture.
- Reactive Forms for timesheet entry.
- NgRx for global state management.
- RxJS for asynchronous data handling.
- Angular Signals for local UI state.
- Role-based routing and UI access.

Deliverables

- Git repository with complete source code.
- README explaining architecture, design patterns, and trade-offs.
- SQL seed data script.
- Unit test execution results.